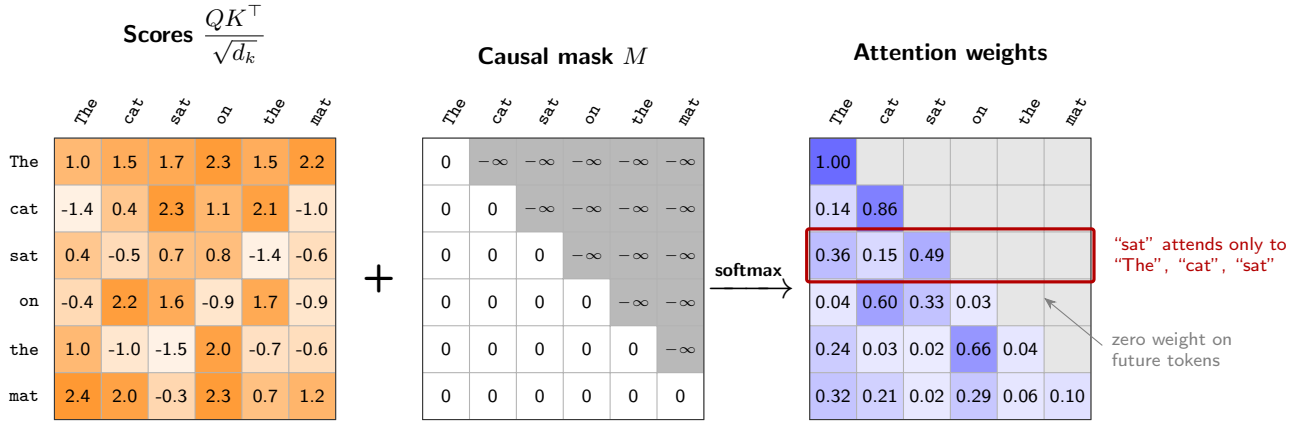


$$\text{CausalAttention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V$$

When is it used?

✓ **Decoder-only LLMs** (GPT-2, GPT-3, Llama): every self-attention layer.
✓ **Decoder of the original Transformer**: masked self-attention.
✓ **Training:** predict all next tokens in parallel without "peeking" at the answer.
✓ **Generation:** each new token depends only on the previous ones.

When is it not used?

× **Encoders** (BERT, ViT): bidirectional, every token sees all tokens.
× **Cross-attention** (decoder → encoder output): the whole source is already known.

The <|pad|> mask is a different mask: it hides padding positions, not the future.